FSL-EGNN: Edge-Labeling Graph Neural Network for Hyperspectral Image Few-Shot Classification

Xibing Zuo, Xuchu Yu, Bing Liu, Pengqiang Zhang, and Xiong Tan

***Abstract*— The existing hyperspectral image (HSI) classification encounters the obstacle of improving the classification accuracy with limited labeled samples. In this context, as a typical implementation of meta-learning, few-shot learning (FSL) makes the model learn by episodic training on source HSI, which has achieved significant improvements in small sample classification of target HSI. However, the existing FSL methods lack explicit consideration and exploration of the association between pixels, especially the intraclass association and interclass association between pixels in the support set and query set. To mitigate these issues, an FSL method based on edge-labeling graph neural network (FSL-EGNN) is proposed for small sample classification of HSI, which is the first attempt to explicitly quantify the associations between pixels by exploiting EGNN in HSI few-shot classification (FSC). Specifically, based on graph construction of HSI, episodic training is performed on the existing source HSI. During training, EGNN is used to predict the edge labels on the graph, thereby explicitly modeling the intraclass similarity and interclass dissimilarity between pixels of HSI. After the trained model is fine-tuned, it can realize FSC on the unseen target HSI. Experiments conducted on three benchmark HSI datasets demonstrate that the proposed FSL-EGNN outperforms the existing state-of-the-art methods with limited labeled samples.**

***Index Terms*— Deep learning, edge-labeling graph neural network (EGNN), few-shot learning (FSL), hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL images (HSIs) typically have a wide range of electromagnetic wave imaging and high spectral resolution, simultaneously, HSI contains rich geospatial information and spectral information, so it has unique advantages in ground object recognition and classification [1], [2]. HSI classification aims to divide each pixel into a unique class [3] based on the inherent spectral characteristics or spatial characteristics of the image. HSI classification has been widely applied in many fields over the past decades, including ecological environment monitoring [4], resource exploration [5], precision agriculture [6]–[8], and so on. However, the characteristics of HSI data such as high dimension, limited training samples, and highly nonlinear structure restrict the fine classification of ground objects.

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The traditional HSI classification methods consist of two separate parts, handcraft feature engineering and classifiers, both of which have generated a lot of pioneering research. The commonly used handcraft feature extraction methods include spectral feature extraction algorithms, such as principal component analysis (PCA) [9] and linear discriminant analysis (LDA) [10], and spatial feature extraction algorithms, such as extended morphological profile (EMP) [11], local binary pattern (LBP) [12], and simple linear iterative cluster (SLIC) [13]. Traditional classifiers mainly include support vector machine (SVM) [14], k-nearest neighbor (KNN) [15], and so on. However, traditional HSI classification methods generally have defects in two aspects. On one hand, their classification principle only focuses on isolated pixel spectral features without exploring the spatial correlation among pixels. On the other hand, parameters' determination of traditional classification methods depends on expert experience and grid search, and the trained model lacks robustness. Therefore, traditional classifiers often generate unsatisfactory classification results with limited training samples.

For the past few years, with the development of big data and the improvement of computing power, deep learning has been extensively used in HSI classification and has drawn increasing attention due to its powerful representation ability [16]. Compared with the traditional classification methods, deep-learning-based methods can obtain high-level features from raw data through a series of stacked layers, and automatic parameter tuning in the process of back propagation enables deep learning to adapt to diversified classification tasks. In deep learning methods, convolutional kernels based on regular grids are first designed and continuously evolved [17]–[19]. Chen *et al.* [20], [21] pioneer the design of stacked autoencoder (SAE) and deep belief network (DBN), revealing the great potential of deep learning methods in the field of HSI classification. Subsequently, convolution neural network (CNN) [22], recurrent neural network (RNN) [23], [24], 2-D CNN (2DCNN) [25]–[29], 3-D CNN (3DCNN) [30]–[32], generative adversarial network (GAN) [33]–[36], capsule network [37]–[39], and transformer network [40], [41] are applied to HSI classification successively. Moreover, to further enhance the quality of HSI classification, residual structure [42], [43], dense connection structure [44], [45], multiscale structure [30], [46], and attention mechanism [47]–[49] have been introduced into the basic structure and achieved exciting performance. However, when only a few numbers of training samples are available, the

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aforementioned supervised classification models are prone to over-fitting, resulting in unsatisfactory classification performance. To this end, several semisupervised classifiers, weakly supervised learning [50], [51], capsule neural network (CapsNet) [52], [53] and unsupervised feature extractors based on deep learning have been effectively explored. Liu *et al.* [54] design a novel semisupervised convolutional neural network named SS-CNN, which uses a large number of unlabeled samples in the training process and improves the classification accuracy of HSI even with a few labeled samples. SS-CNN consists of the clean encoder, the corrupted encoder, and the decoder; skip connections are added between the encoder layer and the decoder layer of SS-CNN; and the sum of supervised loss function and semisupervised loss function is considered in the training process of SS-CNN. Mei *et al.* [55] design an unsupervised spatial–spectral feature extractor named 3-D-CAE, which reconstructs HSI input through 3-D convolution layer and 3-D deconvolution layer, and effective spatial–spectral features can be obtained in an unsupervised manner by comparing with the HSI input; SVM is eventually used to obtain HSI classification results. Jia *et al.* [56] embed Siamese network into semisupervised learning pattern and propose a method called 3DAES, which extracts unsupervised features on a large number of unlabeled samples using autoencoder, and then trains Siamese network with limited training samples to correct unsupervised features to improve separability between classes. Therefore, 3DAES can effectively enhance the classification performance of HSI with limited samples.

Even though the existing CNN-based methods have matured and achieve excellent performance, they still suffer from inevitable shortcomings. Specifically, CNN-based methods adopt various convolution kernels based on regular grids to perform convolution operations on HSI, which makes them insufficient in capturing the global correlation among pixels in HSI. In contrast, the graph neural network (GNN) ignores the inherent grid structure of HSI. It can directly establish connections between graph nodes and neighbor nodes at any location through the edge and can aggregate the information of each neighbor node to generate the characteristics of every graph node itself. Recently, graph neural network has become a hot topic in HSI classification. Qin *et al.* [57] design a semisupervised learning framework based on the spectral–spatial graph convolutional network (S^2GCN), which considers the features of neighboring nodes in the graph and the features of neighboring pixels in the HSI patches neighboring. However, S^2GCN uses a fixed graph, which is prone to overfitting during training. Wan *et al.* [58] propose a multiscale dynamic graph convolution network (MDGCN); this method used SLIC to divide the HSI into many uniform super pixel regions, and then construct graphs at different scales, and implemented dynamic convolution on the graphs. MDGCN effectively overcomes the limitations of S^2GCN. With the deepening of research, apart from the continuous development of GCN [59], [60], graph neural networks such as graph attention network (GAT) [61] and graph sampling and aggregation network (Graphsage) [62] have been successfully developed in HSI classification and achieved fine classification results.

The deep learning models contain thousands of parameters, so a lot of labeled samples are required during the training process. However, in real scenarios, manual labeling of training samples in HSI is a time-consuming and laborious task, so there is an imbalance between labeling samples and the huge parameter space. On the contrary, humans are good at learning through rarely labeled examples. For example, humans only need to see cats a few times before they can easily identify other types of cats. Similarly, few-shot learning (FSL) focuses on learning knowledge that can be generalized to unknown classes from rarely labeled samples. Specifically, after the FSL model has learned numerous data in some classes, the model can be quickly deployed on new classes with limited labeled samples. Recently, FSL has attracted attention in HSI classification. Liu *et al.* [63] propose a deep FSL (DFSL) method, based on the 3-D residual network; DFSL takes the precollected HSI training datasets to learn a metric space, which can be extended to the unknown HSI test datasets to obtain discriminative features, and then, SVM or NN classifier is combined to improve the classification accuracy with a few labeled samples. Gao *et al.* [64] add fine-tuning operation after model training and design a relational network for HSI few-shot classification (RN-FSC). Tong *et al.* [65] proposed an attention-weighted graph convolutional network (AwGCN) for HSI FSC. AwGCN quantifies the relationship between HSI pixels by constructing an attention-weighted graph and applies GCN to propagate information among nodes and obtain node labels. AwGCN is the first method to apply GNN to HSI FSC. Li *et al.* [66] effectively combine FSL and domain adaptation, and they propose a deep cross-domain FSL (DCFSL) method, which considers the domain shift between the source domain and the target domain in FSL.

FSL has been successfully applied to HSI classification. However, the existing FSL methods have not fully explored the association between pixels, especially the intraclass similarity and interclass dissimilarity between the support set and the query set in each task. In view of the fact that the edges of graph can express complex interaction between data nodes explicitly, this article embeds the graph neural network into the FSL framework and proposes an FSL method based on edge-labeling graph neural network (FSL-EGNN). It is worth noting that FSL-EGNN is the first attempt to explicitly quantify the associations between pixels by exploiting EGNN in HSI FSC. Different from the node-labeling framework of AwGCN [65], which implicitly expresses the associations between nodes, FSL-EGNN is an edge-labeling framework that explicitly quantifies the intraclass similarity and interclass dissimilarity between the support set and the query set through edge labels. Specifically, task-based episodic training is performed on the HSI datasets, where the pixels of the support set and the query set together form a fully connected (FC) graph, and the edge features and edge labels in the graph are initialized. In the designed EGNN, node features and edge features are updated iteratively and cyclically, and edge labels can be obtained from the final edge features. Finally, the labels of query set can be predicted by taking edge label as weight and combining with labels of support set. The experimental

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results demonstrate that FSL-EGNN outperforms the existing state-of-the-art methods with limited labeled samples.

The main contributions of this article can be summarized as follows.

1) Different from the general deep learning methods, the FSL method is used to overcome the realistic obstacle of small samples in HSI. Using task-based episodic training strategy, FSL-EGNN is suitable for performing in unseen classes without retraining.
2) It is the first time that EGNN is embedded in the FSL framework in HSI classification; the intraclass similarity and interclass dissimilarity between the pixels of the support set and the query set are explicitly expressed through edge labels. EGNN consists of a stack of layers, and the node features and edge features are cyclically updated through layers to obtain more accurate edge labels. The final edge labels can fully describe the relationship between nodes.
3) The experimental results obtained on three benchmark HSI datasets show that the proposed FSL-EGNN has the potential of classification under limited sample conditions and provides competitive performance advantages compared with the existing state-of-the-art classification methods.

The remainder of this article is structured as follows. Section II describes the definition of FSL and GNN in detail. Section III elaborates on the proposed FSL-EGNN. Extensive experiments and analysis are given in Section IV. Section V concludes this article with some remarks and hints at plausible future research work.

II. Definition of FSL and GNN

A. Few-Shot Learning

FSL is an implementation of meta-learning. HSI FSL aims to learn a classifier that performs well in the case where only a few labeled samples per each class are given in HSI. To carry out FSL, the HSI datasets are divided into the source datasets D_S available for the whole labeled samples and the target datasets D_T to be classified with only a few labeled samples. The number of classes of the source datasets and target datasets is denoted as C_s and C_t, respectively; at the same time, to ensure the diversity of the source datasets, C_s should be larger than C_t. D_T can be further divided into fine-tuning datasets D_f and testing datasets D_t. Generally, D_f and D_t share the same label space, $D_f \cup D_t = D_T$, $D_S \cap D_T = \emptyset$. The process of FSL mainly includes three stages: model training, model fine-tuning, and model testing. To be specific, model training is first carried out on D_S to extract transferable feature knowledge and cultivate the learning ability of the model. Then fine-tune the trained model with D_f to make the model adapt to the classification task on the target dataset quickly. Finally, the trained model is applied to D_t for classification. Note that in the setup of this article, both D_f and D_t only use a few labeled samples in D_T (such as selecting five labeled samples per class), and D_f shares the same labeled samples with D_t.

Different from general deep learning methods that adopt batch-based training methods, FSL adopts task-based episodic training strategies. In other words, each FSC task $\mathcal{T}$ represents an episode, which contains a support set S and a query set Q. The support set serves as a set of labeled samples to improve the generalization ability of the model. The query set is an unlabeled sample set, which is used to calculate the training loss and evaluate the classification ability of the model; the training procedure is iteratively carried out episode by episode until convergence. In the model training phase, each task randomly selects N classes from the source datasets and randomly selects $(K + T)$ samples from each selected class, then the first K samples form the support set, and the last T samples form the query set. Thus, such a situation is called N-way K-shot T-query classification setting. In general, the N-way K-shot T-query setting is formed as follows: $\mathcal{T} = S \cup Q$ where $S = \{\mathbf{x}_i, y_i\}_{i=1}^{N\times K}$ and $Q = \{\mathbf{x}_i, y_i\}_{i=N\times K+1}^{N\times(K+T)}$. Here, $\mathbf{x}_i$ and $y_i \in \{C_1, \ldots, C_N\}$ represent the feature of the ith input data and its corresponding label, respectively.

B. Graph Neural Network

In recent years, with the continuous exploration of deep learning in non-Euclidean domains, GNN has attracted more and more attention and has been widely used in the fields of natural language processing and computer vision. Non-Euclidean data can be represented by a graph structure consisting of nodes and edges. Nodes represent the research object, and edges represent the specific relationship between neighbor nodes. GNN models the features of target nodes by aggregating the feature embeddings of the target nodes and its neighbors. To facilitate the subsequent description, let $\mathcal{G} = (\mathcal{V}, \mathcal{E}; \mathcal{T})$ be the graph constructed with samples from the task $\mathcal{T}$, in which $\mathcal{V} = \{V_i\}_{i=1,2,\ldots,|\mathcal{T}|}$ and $\mathcal{E} = \{E_{ij}\}_{i,j=1,2,\ldots,|\mathcal{T}|}$ denote the nodes and edges of the graph, respectively. $|\mathcal{T}| = N \times (K+T)$ denotes the number of samples in task $\mathcal{T}$. Let $\mathbf{v}_i$ and $\mathbf{e}_{ij}$ be the node feature and the edge feature, respectively.

III. Methodology

This section describes the proposed FSL-EGNN; as illustrated in Fig. 1, the proposed FSL-EGNN consists of four parts: the feature embedding module (FEM), the edge initialization module (EIM), the EGNN, and the label prediction module (LPM). Specifically, taking the two-way two-support one-query task in the model training stage as an example, the given source HSI datasets are first divided into many data cubes containing abundant spectral information and spatial information. For each FSL task, two classes are randomly selected from the data cubes, and three samples are randomly selected for each class. The first two samples are used as the support set, and the last one is used as the query set. Subsequently, the support set and the query set are jointly input into the FEM to extract deep-level recognizable features. Immediately after feature extraction, an FC graph is initially constructed where each node represents each sample, and each edge describes the relationship between the two connected nodes. Then feeding the initialized graph into EGNN, in which the node features and edge features are cyclically updated, and the edge labels in the graph are obtained. Finally, the edge labels serve as the weight, and the labels of the

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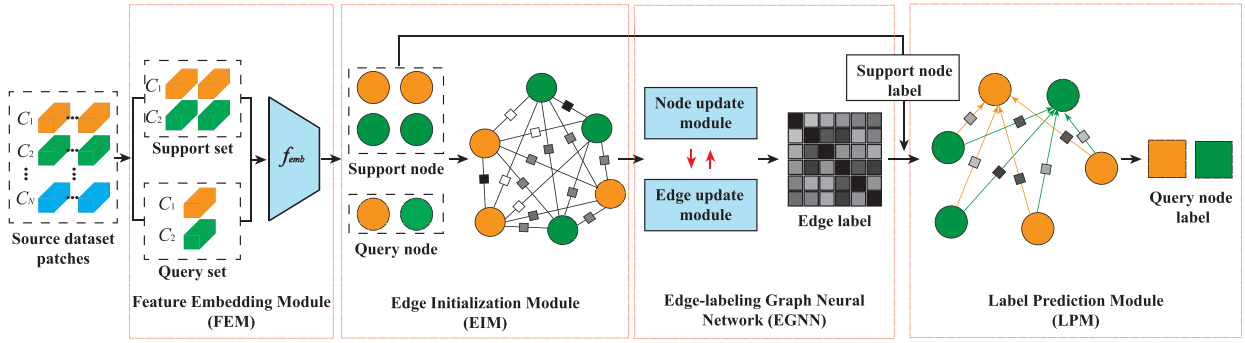


Fig. 1. Framework of the proposed FSL-EGNN. The solid circle represents the node of the support set. The dotted circle represents the node of the query set. The square represents the edge label. The darker the color of the square, the closer the edge label to 1, that is, the greater the similarity within the node class connected by the edge.

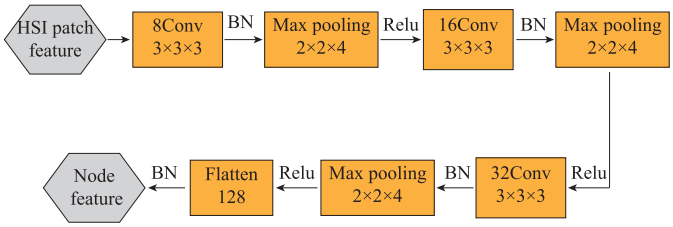


Fig. 2. Architecture of the feature embedding network.

supporting set are used to predict the labels of the query set. In Sections III-A–III-D, the four parts of FSL-EGNN will be presented in detail.

A. Feature Embedding Module

The original HSI contains lots of noise and redundant information. To obtain robust and discriminative spectral–spatial node feature, we design the FEM based on 3-D convolution, as shown in Fig. 2. For the input HSI patches $D \in \mathbb{R}^{d\times m\times n}$, where d, m, n denotes the channels, height, and width of patch, the final node feature is obtained through feature extraction of the three-layer convolution block. Each layer of the convolution block includes a 3-D convolution layer, batch normalization layer, max-pooling layer, and rectified linear unit (ReLU) operation. Compared with 2-D convolution, 3-D convolution can be directly applied to HSI cubes, and it takes into account the relationship between adjacent bands, and therefore it can extract more robust and discriminative spectral–spatial information. We adopt 3-D convolution kernels with a size of $3 \times 3 \times 3$, the number of convolution kernels increases in order of 8, 16, and 32, and the stride of the max-pooling layer is set to $2 \times 2 \times 4$.

The FEM is used to preprocess HSI patch, which not only reduces data redundancy but also extracts discriminative graph node features. Its input and output are HSI patch $\mathbf{x}_i$ and the initial graph node feature $\mathbf{v}_i^0$ of EGNN, respectively. We denote FEM as $\mathbf{v}_i^0 = f_{\text{FEM}}(\mathbf{x}_i, \theta_{\text{fem}})$, where θ_{fem} is the corresponding parameter set.

B. Edge Initialization Module

Graph structure is composed of nodes and edges, and the edges can represent the relationship between graph nodes. If the HSI is constructed into a graph, where each graph node represents a pixel of HSI, then the edges can explicitly model the intracluster similarity and intercluster dissimilarity between connected pixel pairs. Therefore, edge features and edge labels play an important role in characterizing the relationship between two node pairs. The ground truth of each edge label y_{ij} is defined according to whether the ground truth between the node labels is equal, as follows:

$$y_{ij} = \begin{cases} 1, & \text{if } y_i = y_j \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Each edge feature $\mathbf{e}_{ij} = \{e_{ijd}\}_{d=1}^2 \in [0, 1]^2$ is a 2-D vector which denotes the strength of the intraclass and interclass relationship between node pairs, and the intention of this processing is to separate the similarities and dissimilarities between classes. Specifically, the edge features are initialized by edge labels as follows:

$$\mathbf{e}_{ij}^0 = \begin{cases} [1 \parallel 0], & \text{if } y_{ij} = 1 \text{ and } i, j \leqslant N \times K \\ [0 \parallel 1], & \text{if } y_{ij} = 0 \text{ and } i, j \leqslant N \times K \\ [0.5 \parallel 0.5], & \text{otherwise} \end{cases} \tag{2}$$

where $\parallel$ denotes the concatenation operation. The first dimension of the edge feature is used as the edge label $\hat{y}_{ij}$, $\hat{y}_{ij}^0 = e_{ij1}^0$.

EIM is responsible for the initialization of edge features in the graph, whose input and output are the node label pairs (y_i, y_j) and the initial edge features $\mathbf{e}_{ij}^0$, respectively. Consequently, we denote EIM as $\mathbf{e}_{ij}^0 = f_{\text{EIM}}(y_i, y_j)$. Because the initialization of edge features is only related to the ground truth of node label, f_{EIM} has no parameters to be trained.

C. Edge-Labeling Graph Neural Network

In the three stages of FSL, the labels of the query nodes are always unknown. Therefore, (2) initializes the edge features connected to the query node to $[0.5 \parallel 0.5]$. However, the initial edge features cannot fully project the intraclass similarity and interclass dissimilarity between graph nodes. Therefore, we designed EGNN to update the edge features iteratively for estimating the association between query node and support node; Fig. 3 reveals the architecture of EGNN. To be specific, the input of EGNN is the initial graph node features $\mathbf{v}_i^0$ and the initial edge features $\mathbf{e}_{ij}^0$; the node feature update

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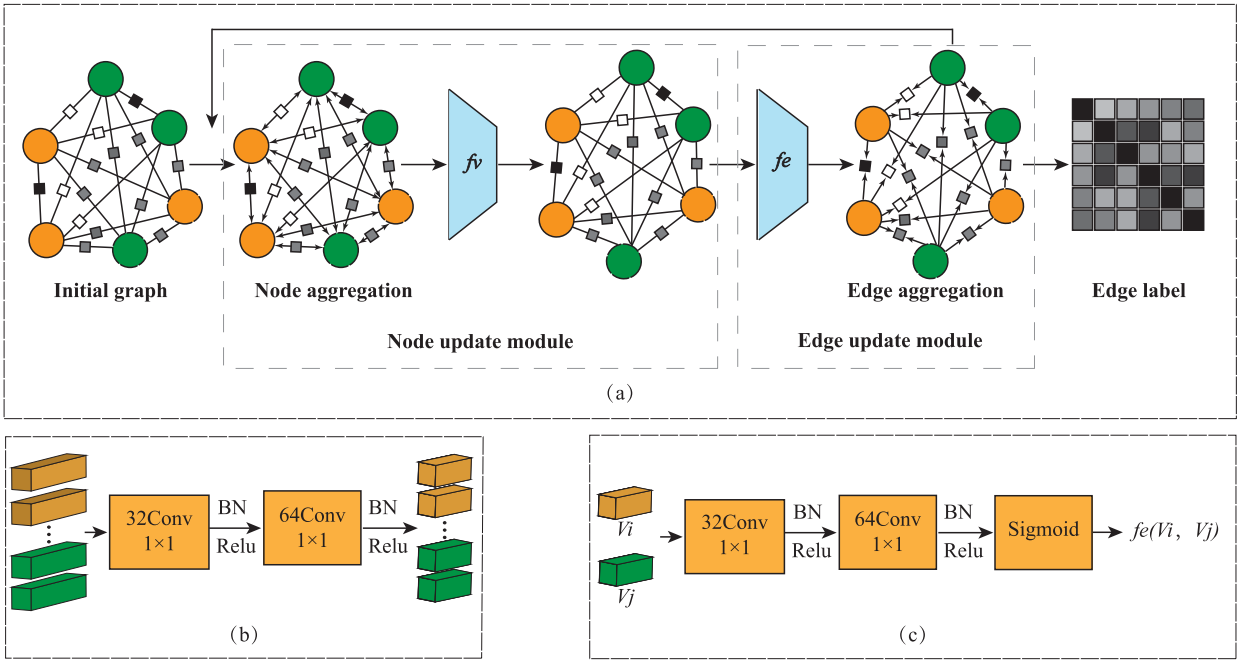


Fig. 3. Architecture of the EGNN model. (a) Overall structure of EGNN. (b) Structure of the node update module. (c) Structure of the edge update module.

module (NUM) and the edge feature update module (EUM) are sequentially implemented through the L layers, and finally the edge label $\hat{y}_{ij}$ is obtained.

1) Node Update Module: As shown in Fig. 3(a), NUM contains two steps: node feature aggregation and node feature transformation. In detail, given $\mathbf{v}_i^{l-1}$ and $\mathbf{e}_{ij}^{l-1}$ from the $l-1$ layer, the feature node $\mathbf{v}_i^l$ is updated by first aggregating feature of adjacent nodes, and then performing the feature transformation. In the process of node feature aggregation, similar to the attention mechanism, the edge feature $\mathbf{e}_{ij}^{l-1}$ at the layer $l-1$ is regarded as the contribution degree of the adjacent node to the target node during aggregation. It is represented as follows:

$$\mathbf{v}_i^l = f_v^l\left(\left[\sum_j^{N\times K+T} \tilde{e}_{ij1}^{l-1}\mathbf{v}_j^{l-1} \,\middle\|\, \sum_j^{N\times K+T} \tilde{e}_{ij2}^{l-1}\mathbf{v}_j^{l-1}\right]; \theta_v^l\right) \tag{3}$$

where $\tilde{e}_{ijd} = (e_{ijd}/\sum_k^{e_{ikd}})$, $d = 1, 2$ denotes the d-dimensional vector value of normalized edge feature, and f_v^l denotes the node feature transformation network with parameter set θ_{num}^l in the layer l. As shown in Fig. 3(b), the feature transformation network consists of 2-D convolution kernels with a size of 1×1, batch normalization, and ReLU operations. It is worth noting that when aggregating node features $\mathbf{v}_i^l$ in (3), both the intraclass similarity $\tilde{e}_{ij1}^{l-1}$ and the interclass dissimilarity $\tilde{e}_{ij2}^{l-1}$ between nodes are considered, which is beneficial to obtain discriminative node features.

2) Edge Update Module: Immediately after NUM, EUM updates the edge features in the graph. The input and output of EUM are the updated node features and the updated edge features, respectively. Specifically, EUM mainly includes two parts: metric network and edge update. The metric network is used to calculate the similarity and dissimilarity between node pairs. Given the previous edge features $\mathbf{e}_{ij}^{l-1}$ and updated node features $\mathbf{v}_i^l$, we can update the edge features as follows:

$$\breve{e}_{ij1}^l = \frac{f_e^l\left(\mathbf{v}_i^l, \mathbf{v}_j^l; \theta_e^l\right) e_{ij1}^{l-1}}{\sum_k f_e^l\left(\mathbf{v}_i^l, \mathbf{v}_k^l, \theta_e^l\right) e_{ik1}^{l-1} / \left(\sum_k e_{ik1}^{l-1}\right)} \tag{4}$$

$$\breve{e}_{ij2}^l = \frac{\left(1 - f_e^l\left(\mathbf{v}_i^l, \mathbf{v}_j^l; \theta_e^l\right)\right) e_{ij2}^{l-1}}{\left(1 - \sum_k f_e^l\left(\mathbf{v}_i^l, \mathbf{v}_k^l, \theta_e^l\right)\right) e_{ik2}^{l-1} / \left(\sum_k e_{ik2}^{l-1}\right)} \tag{5}$$

$$\mathbf{e}_{ij}^l = \breve{\mathbf{e}}_{ij}^l / \left\|\breve{\mathbf{e}}_{ij}^l\right\|_1 \tag{6}$$

where f_e^l denotes the metric network which calculates the similarity scores with parameter θ_e^l. The structure of the metric network is shown in Fig. 3 (c). On the premise of taking the features of node pairs as input, the metric network contains two 2-D convolutional layers with a convolution kernel size of 1×1 and the number of convolution kernels in order of 32 and 64. Each 2-D convolutional layer is followed by a batch normalization and ReLU nonlinearity activation function. The final activation function is sigmoid, which is adopted to obtain the final similarity score.

In conclusion, each layer of EGNN contains NUM and EUM. For the lth layer in EGNN, NUM first uses the node features $\mathbf{v}_i^{l-1}$ and edge features $\mathbf{e}_{ij}^{l-1}$ of the $(l-1)$th layer to obtain the node features of the lth layer $\mathbf{v}_i^l$. Subsequently, EUM uses the node features $\mathbf{v}_i^l$ and the edge features $\mathbf{e}_{ij}^{l-1}$ to update $\mathbf{e}_{ij}^l$. Both $\mathbf{v}_i^l$ and $\mathbf{e}_{ij}^l$ can be used as the input of the $(l+1)$th layer in EGNN. After L iterations to update node features and edge features, the predicted edge labels can be obtained from the final edge feature $\mathbf{e}_{ij}^L$, that is, $\hat{y}_{ij} = \mathbf{e}_{ij1}^L$. Here, $\hat{y}_{ij} \in [0, 1]$. In general, we denote EGNN as $\hat{y}_{ij} = f_{\text{EGNN}}(\mathbf{v}_i^0, \mathbf{e}_{ij}^0, \theta_v^l, \theta_e^l)$. θ_v^l and θ_e^l represent the trainable parameters of NUM and EUM in the lth layer, respectively.

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D. Label Prediction Module

After EGNN obtains the predicted edge labels, we should consider how to predict the labels of query nodes. Since the first dimension and the second dimension of the edge feature, respectively, represent the intraclass similarity and the interclass dissimilarity between nodes, $\hat{y}_{ij} = \mathbf{e}_{ij1}^{L}$, $\hat{y}_{ij} \in [0, 1]$ can be regarded as the probability that two nodes come from the same class. Therefore, based on the known support node labels, the label of each query node $\hat{y}_i$ can be determined by voting with $\hat{y}_{ij}$ as the weight. The specific expression is as follows:

$$\hat{y}_i = \operatorname{argmax}(p_i(n)) \tag{7}$$

where argmax is a function representing the maximum index of $p_i(n)$, and $p_i(n)$ denotes the probability that node i belongs to class n, which is defined as

$$\begin{aligned} p_i(n) &= p(y_i = C_n \mid \mathcal{T}) \\ &= \operatorname{softmax}\left(\sum_{\{(x_j, y_j) \in S\}} \hat{y}_{ij} \delta\big(y_j = C_n\big) \right) \end{aligned} \tag{8}$$

where $\delta(y_j = C_n)$ is the Kronecker delta function defined as follows:

$$\delta\big(y_j = C_n\big) = \begin{cases} 1, & \text{if } y_j = C_n \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

LPM takes the predicted edge labels $\hat{y}_{ij}$ as input and node predicted label $\hat{y}_i$ in the query set as output. Therefore, LPM can be denoted as $\hat{y}_i = f_{\mathrm{LPM}}(\hat{y}_{ij})$. To sum up, the whole FSL-EGNN can be denoted as follows:

$$\hat{y}_i = f_{\mathrm{LPM}}\big(f_{\mathrm{EGNN}}\big(f_{\mathrm{FEM}}(\mathbf{x}_i, \theta_{\mathrm{fem}}), f_{\mathrm{EIM}}\big(y_i, y_j\big), \theta_v^l, \theta_e^l\big)\big). \tag{10}$$

The implementation procedure of FSL-EGNN is summarized in Algorithm 1. During FSL, the parameters of the proposed FSL-EGNN, $\{\theta_{\mathrm{fem}}, \theta_v^l, \theta_v^l\}_{l=1}^{L}$, are trained in an end-to-end manner by minimizing the following loss function:

$$\mathcal{L} = \sum_{l=1}^{L} \mathcal{L}_e\big(Y_e, \hat{Y}_e^l\big) \tag{11}$$

where Y_e and $\hat{Y}_e^l$ are the ground-truth query edge labels and the predicted query edge labels at the lth layer of EGNN, respectively. The edge loss $\mathcal{L}_e$ is defined as binary cross-entropy loss. In addition, to improve the gradient descent of the lower layers, we combine the edge loss of all L layers of EGNN to obtain the total loss.

Algorithm 1 Implementation Procedure of FSL-EGNN

Input: HSI data patches $D \in \mathbb{R}^{d \times m \times n}$, number of randomly sampled classes N, number of support samples per class K, number of query samples per class T.
Output: The predicted labels $\{\hat{y}_i\}_{i=N\times K+1}^{N\times(K+T)}$.
Initialization : $\mathcal{T} = S \cup Q$, $S = \{\mathbf{x}_i, y_i\}_{i=1}^{N\times K}$, $Q = \{\mathbf{x}_i, y_i\}_{i=N\times K+1}^{N\times(K+T)}$, $y_i \in \{C_1, \ldots, C_N\}$.
▶ Feature embedding module:
  $\mathbf{v}_i^0 = f_{FEM}(\mathbf{x}_i, \theta_{fem})$
▶ Edge initialization module:
  $\mathbf{e}_{ij}^0 = f_{EIM}(y_i, y_j)$
▶ Edge-labeling graph neural network:
for l in $1, \ldots, L$ **do**
  ▶ Node feature update module:
    Update $\mathbf{v}_i^l$ by Eq. 3
  ▶ Node feature update module:
    Update $\mathbf{e}_{ij}^l$ by Eq. 4, Eq. 5 and Eq. 6
end for
▶ Label prediction module:
  obtain $\hat{y}_i$ by Eq. 7, Eq. 8 and Eq. 9
Update loss $\mathcal{L}$ by Eq. 11

IV. Experiments and Analysis

A. Hyperspectral Datasets Description

1) Source Datasets: To conduct training procedures and mine transferable knowledge in FSL-EGNN, five public datasets from different regions and covering different spatial resolutions are considered as source datasets, including the Houston2013, the Botswana, the Kennedy Space Center (KSC), the Chikusei, and the Pavia Center. The details of the five source datasets are presented in Table I, in which each dataset has six attributes, including "spatial size" (abbreviated as size), "spectral range" (abbreviated as range), "number of bands" (abbreviated as bands), "ground sample distance" (abbreviated as GSD), "sensor type" (abbreviated as sensor), and "number of classes" (abbreviated as Classes). These five source datasets are acquired by different sensors and have different GSD, which ensures the diversity of samples during the training process. There are a total of 70 classes in the source datasets; inspired by Liu *et al.* [63], to ensure the same number of samples for each training class, the classes with fewer than 200 samples were discarded in the source datasets. Therefore, the remaining 64 classes are available for training and 200 samples are randomly selected for each class.

2) Target Datasets: Three well-known HSI datasets are adopted as target datasets to assess the classification performance of our proposed FSL-EGNN, both quantitatively and qualitatively.

a) University of Pavia Dataset: The University of Pavia dataset (abbreviated as UP) was acquired by the reflective optics system imaging spectrometer (ROSIS) sensor over the Pavia University in northern Italy. The spatial size of UP is 610 × 340 pixels, and the corresponding GSD is 1.3 m. This sensor can acquire 115 bands in total in the wavelength range of 430–860 nm. After removing the greatly noise-affected channels, the remaining 103 spectral bands are used for experiments. There are nine mutually exclusive classes contained in the UP dataset; the detailed land-cover classes, number of training samples, and number of testing samples are shown in Table II.

b) Indian Pines Dataset: The Indian Pines dataset (abbreviated as IP) was acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over the northwestern Indiana, USA. The spatial size of IP is 145 × 145 pixels, and the corresponding GSD is 20 m. This sensor can acquire 220 bands in total in the wavelength range of

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TABLE I

DETAILS OF FIVE SOURCE DATASETS, INCLUDING SPATIAL SIZE (SIZE), SPECTRAL RANGE (RANGE), NUMBER OF BANDS (BANDS), GSD, SENSOR TYPE (SENSOR), NUMBER OF CLASSES (CLASSES)

	Size	Range	Bands	GSD	Sensor	Classes
Houston2013	349×1905	380-1050	144	2.5	ITRES-CASI 1500	15
Botswana	1476×256	400-2500	145	30	EO-1	14
KSC	614×512	400-2500	176	18	AVIRIS	13
Chikusei	2517×2335	363-1018	128	2.5	Hyperspec -VNIR-C	19
Pavia Center	1096×715	430-860	102	1.3	ROSIS	9

TABLE II

LAND-COVER CLASSES, NUMBER OF TRAINING SAMPLES, AND NUMBER OF TESTING SAMPLES FOR THE UP DATASET

Class No.	Class Name	Training	Testing
1	Asphalt	5	6631
2	Meadows	5	18649
3	Gravel	5	2099
4	Trees	5	3064
5	Painted metal sheets	5	1345
6	Bare Soil	5	5029
7	Bitumen	5	1330
8	Self-Blocking Bricks	5	3682
9	Shadows	5	947
	Total	45	42776

400–2500 nm. After removing the greatly noise-affected channels, the remaining 200 spectral bands are used for experiments. There are 16 mutually exclusive classes contained in the IP dataset; the detailed land-cover classes, number of training samples, and number of testing samples are shown in Table III.

c) Salinas Dataset: The Salinas dataset (abbreviated as SA) was acquired by the AVIRIS sensor over the region of Salinas Valley, CA, USA. The spatial size of SA is 512×217 pixels, and the corresponding GSD is 3.7 m. This sensor can acquire 224 bands in total in the wavelength range of 400–2500 nm. After removing the greatly noise-affected channels, the remaining 204 spectral bands are used for experiments. There are 16 mutually exclusive classes contained in the SA dataset; the detailed land-cover classes, number of training samples, and number of testing samples are shown in Table IV.

To reduce data redundancy and ensure the same input dimensions of FEM, inspired by Gao *et al.* [64], 100 bands are selected on each source dataset via the graph-representation-based band selection (GRBS) [67] for training. GRBS is an unsupervised band selection method; the band index selected via GRBS on the five source datasets and three target datasets are shown in Table V. Simultaneously, the $P \times P$-neighborhood data cube of the center pixel is considered as the input of FEM. In general, the input size of FEM is $P \times P \times 100$, where P will be discussed in the subsequent experiments.

TABLE III

LAND-COVER CLASSES, NUMBER OF TRAINING SAMPLES, AND NUMBER OF TESTING SAMPLES FOR THE IP DATASET

Class No.	Class Name	Training	Testing
1	Alfalfa	5	46
2	Corn-notill	5	1428
3	Corn-mintill	5	830
4	Corn	5	237
5	Grass-pasture	5	483
6	Grass-trees	5	730
7	Grass-pasture-mowed	5	28
8	Hay-windrowed	5	478
9	Oats	5	20
10	Soybean-notill	5	972
11	Soybean-mintill	5	2455
12	Soybean-clean	5	593
13	Wheat	5	205
14	Woods	5	1265
15	Buildings-Grass-Trees-Drives	5	386
16	Stone-Steel-Towers	5	93
	Total	80	10249

TABLE IV

LAND-COVER CLASSES, NUMBER OF TRAINING SAMPLES, AND NUMBER OF TESTING SAMPLES FOR THE SALINAS DATASET

Class No.	Class Name	Training	Testing
1	Brocoli_green_weeds_1	5	2009
2	Brocoli_green_weeds_2	5	3726
3	Fallow	5	1976
4	Fallow_rough_plow	5	1394
5	Fallow_smooth	5	2678
6	Stubble	5	3959
7	Celery	5	3579
8	Grapes_untrained	5	11271
9	Soil_vinyard_develop	5	6203
10	Corn_senesced_green_weeds	5	3278
11	Lettuce_romaine_4wk	5	1068
12	Lettuce_romaine_5wk	5	1927
13	Lettuce_romaine_6wk	5	916
14	Lettuce_romaine_7wk	5	1070
15	Vinyard_untrained	5	7268
16	Vinyard_vertical_trellis	5	1807
	Total	80	54129

B. Experimental Setup

1) Implementation Details: In our experiment, the whole comparative experiments are carried out on a laptop with an AMD Ryzen 9 5900HX, 3.30 GHz, an Nvidia GeForce RTX 3080Ti GPU, and an 32-GB RAM. The proposed FSL-EGNN is mainly implemented using the pytorch platform and Python language, and Adam is used to optimize the networks, where the weight decay is 10^{-6}. In the process of model training and model fine-tuning, the number of epochs is set to 10 000 and 1000, respectively. The learning

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TABLE V
INDEX OF SELECTED BANDS VIA GRBS ON THE FIVE SOURCE DATASETS AND THREE TARGET DATASETS

HSI data set	Index of the selected bands
Houston2013	2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 77 107 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 132 133 134 135 143 144
Botswana	2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 88 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 137 138 139 140 141 142 143 144 145
KSC	2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 28 29 31 32 33 35 36 37 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 95 101 120 132 143 144 145 146 147 148 149 150 151 155 167 175 176
Chikusei	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 65 66 67 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 116 117 118 119 120 121 122 123 124 125 126 127 128
Pavia Center	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102
Indian Pines	4 5 6 7 8 10 11 12 13 16 17 18 19 20 21 22 24 25 26 27 28 29 31 33 34 35 39 40 41 42 43 44 45 46 48 49 50 51 52 53 54 55 56 57 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 111 115 117 126 151 152 162 176 178 181 185 186 191 195 200
The University of Pavia	2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103
Salinas	2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 31 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 126 139 204

rate can be dynamically updated by cutting in half every 100 epochs.

Furthermore, four widespread classification evaluation coefficients, i.e., class-specific accuracy, overall accuracy (OA), average accuracy (AA), and kappa coefficient ($\mathcal{K}$), are used to assess the classification performance quantitatively. We also exploit classification maps to qualitatively evaluate the experimental results. To improve the reliability and credibility of comparison experiment results, on the basis of randomly selecting training samples, we conduct each classification experiment ten times to calculate the average result and variance.

2) Comparison With State-of-the-Art Baseline Methods: To evaluate the classification performance of the proposed FSL-EGNN, several state-of-the-art methods have been selected for comparative experiments, including SVM with radial basis function kernel (SVM-RBF) [14], spatial–spectral residual network (SSRN) [42], CEGCN [68], DFSL + NN [63], DFSL + SVM [63], RN-FSC [64], and DCFSL [66]. To ensure fair implementation of comparative experiments, identical training samples are used for all the methods. The parameter settings of these methods are described in the following.

a) SVM-RBF: The well-known machine learning library called scikit-learn is used for implementation in our case. The SVM-RBF contains two hyperparameters, namely, the optimal regularization parameter C and the kernel parameter γ, which can be determined by twofold cross validation in the range of $C = \{2^{-2}, 2^{-1}, \ldots, 2^{7}, 2^{8}\}$ and $\gamma = \{2^{-2}, 2^{-1}, \ldots, 2^{7}, 2^{8}\}$, respectively. Moreover, the whole spectral bands of HSI are used as the input of SVM-RBF.

b) SSRN: SSRN is an end-to-end SSRN, which combines 3-D convolution and residual structure and designs spectral residual module and spatial residual module, respectively, to extract discriminative HSI features. The network structure of SSRN mainly includes two spectral residual modules, two spatial residual modules, an average pooling layer, and a fully connected layer. The input of SSRN is $9 \times 9 \times B$ image block, where B denotes the whole spectral bands of the HSI to be classified.

c) CEGCN: CEGCN is a novel hybrid deep learning framework that performs pixel-level and superpixel-level feature fusion by integrating CNN with GCN. The network structure of CEGCN mainly consists of four parts, a superpixel-based graph constructor that uses SLIC to reduce the number of graph nodes, a spectral transformation subnetwork for enhancing feature discrimination, a superpixel-level graph subnetwork including graph encoder, GCN, and graph decoder, and a pixel-level convolutional subnetwork for extracting local spectral–spatial features.

d) DFSL + NN: DFSL + NN is an FSL method for HSI classification. Res-3-D-CNN is used as a network architecture to learn a metric space, in which samples belonging to the same class are close and others from different classes are far away from each other. Subsequently, the trained Res-3-D-CNN is used to extract the features of all the samples on the target dataset, and the obtained discriminative features are input into the nearest neighbor (NN) classifier for classification. The structure of Res-3-D-CNN in DFSL consists of seven 3-D convolutional layers and two max-pooling layers, and the input size is $9 \times 9 \times 100$.

e) DFSL + SVM: DFSL + SVM uses the same FSL method as DFSL + NN. The only difference between the two is the use of different classifiers Specifically, DFSL + SVM uses SVM for HSI classification. The setting of the network structure and the input size of the model are consistent with DFSL + NN.

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f) RN-FSC: RN-FSC is another well-known FSL method, which is similar to the proposed method in terms of learning procedure. The process of RN-FSC consists of the meta-learning stage on the source datasets, the fine-tuning stage, and the classification stage on the target datasets. The RN-FSC model mainly contains a feature learning module and a relationship learning module. These two modules adopt 3-D convolution and 2-D convolution for feature learning. $9 \times 9 \times 100$ data cubes are used as model input.

g) DCFSL: DCFSL takes into account the domain shift between the source domain and the target domain in FSL, which is a novel DCFSL method. DCFSL mainly consists of two mapping layers, a spectral–spatial feature extraction network, and a domain distribution alignment layer. Specifically, based on two mapping layers and a spectral–spatial feature extraction network, FSL is performed simultaneously on the source classes and target classes to learn a discriminative embedding space. Subsequently, in the domain distribution alignment layer, a conditional adversarial domain adaptation strategy is adopted to achieve the alignment of the source domains and target domains. In DCFSL, the neighborhood size of the input HSI is 9×9. Since DCFSL is essentially a transfer learning, the Chikuse dataset is selected as the source domain dataset with reference to the original article.

The general network configuration in each module of FSL-EGNN is shown in Table VI, which details the network structure and the corresponding size of each module.

TABLE VI

GENERAL NETWORK CONFIGURATION IN EACH MODULE OF FSL-EGNN. CONV, BN, MAXPOOLING, FC, AND SIGMOID STAND FOR THE CONVOLUTION, BATCH NORMALIZATION, MAX-POOLING, FULLY CONNECTED LAYER, AND SIGMOID FUNCTION, RESPECTIVELY. P DENOTES THE NEIGHBORHOOD SIZE OF THE DATA CUBE

FSL-EGNN			Size
Input			$P \times P \times 100$
FEM	Block1	Conv1	$3 \times 3 \times 3$ (8)
		BN1	N/A
		Maxpooling1	$2 \times 2 \times 4$
		Relu1	N/A
	Block2	Conv2	$3 \times 3 \times 3$ (16)
		BN2	N/A
		Maxpooling2	$2 \times 2 \times 4$
		Relu2	N/A
	Block3	Conv3	$3 \times 3 \times 3$ (32)
		BN3	N/A
		Maxpooling3	$2 \times 2 \times 4$
		Relu3	N/A
		FC	128
		BN4	N/A
EIM			N/A
EGNN	Layer1(NUM)	Conv1_1	1×1 (32)
		BN1_1+Relu1_1	N/A
		Conv1_2	1×1 (64)
		BN1_2+Relu1_2	N/A
	Layer1(EUM)	Conv1_3	1×1 (32)
		BN1_3+Relu1_3	N/A
		Conv1_4	1×1 (64)
		BN1_4+Relu1_4	N/A
		Sigmoid1_1	N/A
	Layer2(NUM)	Conv2_1	1×1 (32)
		BN2_1+Relu2_1	N/A
		Conv2_2	1×1 (64)
		BN2_2+Relu2_2	N/A
	Layer2(EUM)	Conv2_3	1×1 (32)
		BN2_3+Relu2_3	N/A
		Conv2_4	1×1 (64)
		BN2_4+Relu2_4	N/A
		Sigmoid2_1	N/A
	Layer3(NUM)	Conv3_1	1×1 (32)
		BN3_1+Relu3_1	N/A
		Conv3_2	1×1 (64)
		BN3_2+Relu3_2	N/A
	Layer3(EUM)	Conv3_3	1×1 (32)
		BN3_3+Relu3_3	N/A
		Conv3_4	1×1 (64)
		BN3_4+Relu3_4	N/A
		Sigmoid3_1	N/A
LPM			N/A
Output ($\hat{y}_i$)			1

C. Parameters' Analysis and Setting

Generally, the classification performance of deep learning methods is determined by the quality of hyperparameters. Therefore, it is very important to determine the values of main hyperparameters in the proposed FSL-EGNN. The hyperparameters studied mainly include neighborhood size P, learning rate, EGNN depth L, and episodic training settings (N way K shot T query settings).

1) Analysis of Neighborhood Size: To jointly use spatial and spectral information for HSI classification, the input of FSL-EGNN is defined as many 3-D $P \times P \times 100$ data cubes, where the neighborhood size P determines the size of the spatial neighborhood of each central pixel. Therefore, the neighborhood size P can affect the final classification result. To specifically analyze the impact of neighborhood size, the value of P is set as 9, 11, 13, 15, 17, 19, 21, 23, and 25, respectively. The overall classification accuracy corresponding to different P is shown in Fig. 4. It can be seen from the trend of classification results that small neighborhood size will lead to poor classification performance; when the neighborhood size increases to a certain value (i.e., 17 or 19), the curves of the classification results basically reach saturation and no longer continue to grow. The optimal neighborhood size corresponding to the UP, IP, and Salinas datasets are 19, 21, and 23, respectively. The OA of three datasets changes slightly when the neighborhood size is set to 19, 21, or 23, and the smaller neighborhood size corresponds to less memory occupancy and higher computational efficiency. To ensure that FSL-EGNN is only trained once on the source datasets, the neighborhood size P is uniformly set to 19. Consequently, the input dimension of FSL-EGNN is unified as $19 \times 19 \times 100$.

2) Analysis of Learning Rate: The learning rate, as an important hyperparameter in deep learning, determines whether the loss function can converge to a local minimum

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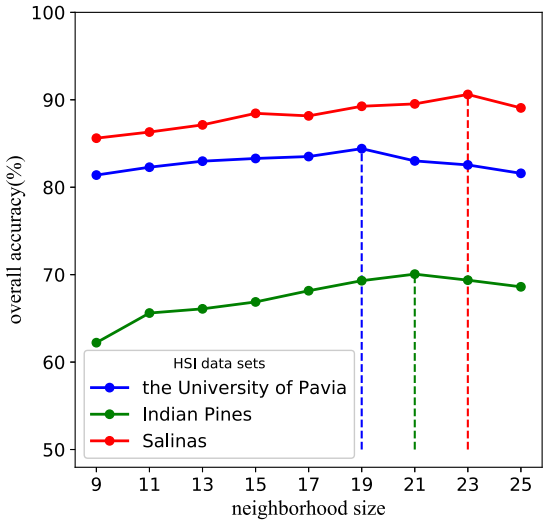


Fig. 4. OA with different neighborhood sizes.

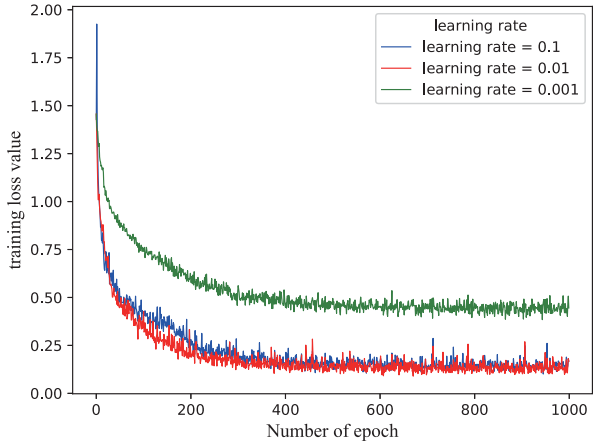


Fig. 5. Training loss values with different learning rates.

and when to converge to the minimum during the training process. To study the effect of learning rate on classification performance, the learning rate is set to 0.1, 0.01, and 0.001, respectively. The training loss values with different learning rates are shown in Fig. 5. From Fig. 5, we can find that a larger learning rate is not conducive to the convergence of the loss function. When the learning rate is set to 0.01, the convergence effect of the loss function is the best. Therefore, the final learning rate is determined to be 0.01.

3) Influence of EGNN Depth: To analyze the influence of EGNN depth L on classification performance, we set the number of network depth to $L = 1$, $L = 2$, $L = 3$, $L = 4$, and $L = 5$, respectively, and calculate the corresponding OA and training time under different values of L, as shown in Fig. 6. From the results in Fig. 6, we can find that shallow network depths (i.e., $L = 1$ or $L = 2$) correspond to poor classification performance, because shallow layers cannot sufficiently extract node information and edge features in the graphs. When the depth of the network is greater than three layers, the OA rises

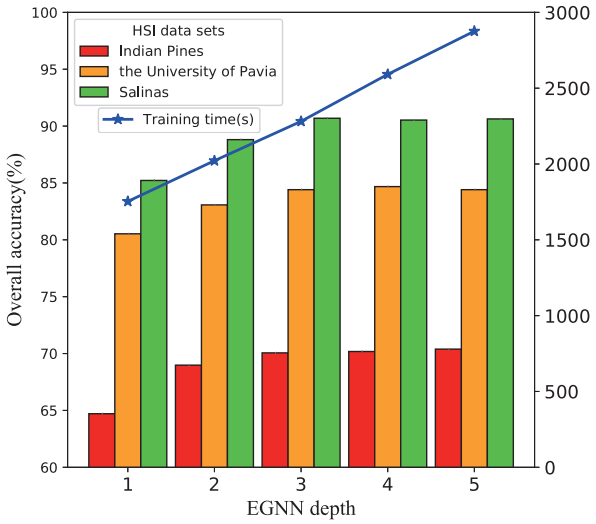


Fig. 6. OA and training time with different network depths.

slowly, but the training time increases sharply. Considering the training time and classification performance comprehensively, $L = 3$ is finally determined. Specifically, the node update module and the edge update module cyclically interact three times.

4) Analysis of Episodic Training Settings: Episodic training promotes model learning to learn in the form of tasks. The parameters contained in the episodic training settings mainly include the number of classes N randomly selected from each task, the number of support set K, and the number of query set T randomly selected from each class, namely, N-way K-shot T-query settings. N determines the diversity of the task in the training procedure, and the source dataset has 64 available classes. Therefore, the values of N are set to 10, 20, 30, 40, 50, and 60, respectively, Fig. 7(a) shows the OA corresponding to different values of N. From the classification results, we can find that when $N = 30$, the classification accuracy is the highest. When the value of N is small or large, the model cannot adapt to simpler or more complex training tasks, so the classification performance is not ideal. Therefore, the optimal value of N is 30. Furthermore, K and T determine the number of samples and the size of each task, respectively. Inspired by Gao *et al.* [64], we fix the size of each task to 20 samples per class and calculate the classification accuracy under different combinations of K and T. The specific experimental settings and experimental results are shown in Fig. 7(b). It can be found that when $K = 1$, the classification accuracy is the highest, and as K increases, the classification accuracy gradually decreases. This also demonstrates that the smaller the number of samples in the support set, the more it can simulate the FSL in the actual HSI classification situation. Consequently, the settings of episodic training are 30-way 1-shot 19-query.

D. Quantitative Evaluation

Tables VII–IX report the classification scores obtained by different methods for the UP, IP, and Salinas datasets,

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TABLE VII

QUANTITATIVE COMPARISON OF DIFFERENT METHODS IN TERMS OF CLASS-SPECIFIC ACCURACY, OA, AA, AND κ ON THE UP DATASET. THE BOLD VALUE REPRESENTS THE BEST ONE IN THE CORRESPONDING ROW

Class No.	SVM-RBF	SSRN	CEGCN	DFSL+NN	DFSL+SVM	RN-FSC	DCFSL	FSL-EGNN
1	**91.85**	91.84	64.71	88.53	88.9	84.93	82.20	88.40
2	84.01	**95.13**	81.94	93.83	94.28	89.44	87.74	84.61
3	29.72	55.23	**86.38**	60.22	69.32	74.23	67.46	79.44
4	51.51	78.02	89.03	76.32	69.05	88.76	**93.16**	85.55
5	93.90	98.34	99.68	**99.82**	99.76	97.73	99.49	99.62
6	36.54	53.56	71.57	57.04	58.06	60.59	77.32	**78.32**
7	35.17	60.07	**99.15**	40.00	41.69	41.18	81.18	92.50
8	62.04	85.34	82.03	62.60	65.63	68.53	66.73	**87.32**
9	**99.79**	98.08	97.72	94.57	90.95	95.60	98.66	96.03
OA (%)	61.35 ±7.11	76.26 ±5.78	80.22 ±4.02	74.23 ±5.96	73.58 ±7.40	77.58 ±8.93	83.65 ±1.77	**85.48** ±2.47
AA (%)	64.95 ±2.99	79.51 ±3.21	85.80 ±1.52	74.77 ±3.73	75.29 ±4.36	77.89 ±3.71	83.77 ±1.74	**87.98** ±2.01
κ ×100	52.71 ±7.07	70.56 ±6.69	74.63 ±4.56	67.69 ±6.75	67.23 ±8.21	71.29 ±10.03	78.70 ±2.01	**81.11** ±3.19

TABLE VIII

QUANTITATIVE COMPARISON OF DIFFERENT METHODS IN TERMS OF CLASS-SPECIFIC ACCURACY, OA, AA, AND κ ON THE IP DATASET. THE BOLD VALUE REPRESENTS THE BEST ONE IN THE CORRESPONDING ROW

Class No.	SVM-RBF	SSRN	CEGCN	DFSL+NN	DFSL+SVM	RN-FSC	DCFSL	FSL-EGNN
1	27.45	18.38	88.04	64.45	30.71	58.41	95.37	**98.26**
2	37.9	**64.79**	46.09	56.47	51.48	45.85	43.26	58.11
3	24.13	27.65	48.10	47.95	35.37	31.32	**57.95**	56.47
4	25.04	26.97	54.60	68.84	32.93	24.85	80.60	**82.95**
5	54.59	80.76	76.48	43.65	64.81	**89.49**	72.91	76.23
6	81.61	86.87	**97.97**	69.18	82.73	93.48	87.96	91.26
7	26.46	32.24	**100.00**	51.24	48.15	15.75	99.57	**100.00**
8	94.59	**100.00**	93.26	73.90	97.74	99.55	86.26	92.91
9	12.74	57.69	99.50	42.69	32.44	13.89	99.33	**100.00**
10	36.58	59.69	**69.56**	41.96	44.06	27.51	62.44	57.60
11	59.51	**70.87**	55.68	65.59	67.74	49.73	62.75	67.56
12	21.55	45.00	39.11	51.51	39.04	22.81	48.72	**52.01**
13	71.75	88.29	**99.66**	66.67	89.74	94.69	99.35	96.98
14	86.02	97.18	87.74	92.94	**93.59**	93.15	85.40	92.70
15	29.60	36.64	70.47	**84.87**	58.51	52.57	66.69	72.64
16	80.59	60.98	**99.89**	97.04	65.26	65.82	97.61	98.17
OA (%)	47.86 ±2.04	61.36 ±0.49	65.95 ±2.77	60.04 ±1.19	59.44 ±1.27	52.49 ±4.09	66.81 ±2.37	**67.99** ±1.88
AA (%)	48.13 ±2.88	59.75 ±0.20	76.63 ±1.68	63.68 ±1.49	58.39 ±2.30	54.93 ±1.27	77.89 ±0.86	**80.87** ±1.80
κ ×100	41.75 ±2.08	56.91 ±0.48	61.52 ±2.80	54.93 ±1.21	54.65 ±1.38	47.30 ±3.98	62.64 ±0.86	**63.71** ±2.16

respectively. Note that the bold parts in these tables represent the optimal value for each row. From the classification results in these tables, we can draw several conclusions as listed below.

1) In the case of consistent training samples, compared with the traditional machine learning classifier (i.e., SVM-RBF), deep learning methods (i.e., SSRN, CEGCN, DFSL + NN, DFSL + SVM, RN-FSC, DCFSL, and FSL-EGNN) can achieve better classification performance most of the time. This is mainly because the deep network can effectively extract the discriminative spectral and spatial information of HSIs, yielding higher classification accuracies. Among all the contrasting methods, CEGCN effectively combines CNN with GCN,

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TABLE IX

QUANTITATIVE COMPARISON OF DIFFERENT METHODS IN TERMS OF CLASS-SPECIFIC ACCURACY, OA, AA, AND κ ON THE SALINAS DATASET. THE BOLD VALUE REPRESENTS THE BEST ONE IN THE CORRESPONDING ROW

Class No.	SVM-RBF	SSRN	CEGCN	DFSL+NN	DFSL+SVM	RN-FSC	DCFSL	FSL-EGNN
1	92.88	97.55	99.89	98.54	96.52	**100.00**	99.40	97.79
2	98.3	98.97	98.54	96.68	95.83	97.53	99.76	**99.93**
3	80.47	92.47	**99.96**	88.93	89.07	88.75	91.96	90.53
4	96.83	96.50	**99.61**	98.42	98.59	94.25	99.55	98.70
5	89.04	94.20	97.11	**98.81**	96.81	97.52	92.70	94.84
6	99.79	99.28	99.90	99.90	**99.95**	99.65	99.52	99.78
7	90.47	99.98	99.94	96.42	91.77	**99.99**	98.88	99.78
8	67.42	**86.90**	75.17	83.98	78.97	72.49	74.57	76.75
9	96.45	99.64	**99.93**	97.52	94.89	99.14	99.59	99.12
10	77.64	**92.01**	83.63	91.01	86.26	91.06	86.42	88.34
11	57.96	95.86	**98.55**	65.44	53.96	67.41	96.61	97.18
12	91.73	99.15	**99.99**	91.26	93.94	98.94	99.93	98.83
13	83.27	89.24	**99.86**	61.15	61.44	98.07	99.30	99.72
14	83.99	95.15	98.23	86.65	87.64	96.97	98.85	**99.10**
15	51.21	55.97	64.70	64.49	61.42	51.38	75.38	**78.27**
16	90.32	**98.91**	93.97	78.56	71.58	92.32	92.22	94.84
OA (%)	79.27 ±3.42	86.39 ±2.68	88.55 ±3.65	85.85 ±1.36	82.51 ±2.69	83.89 ±3.23	89.34 ±2.19	**90.39** ±2.27
AA (%)	84.23 ±1.96	93.24 ±1.29	94.31 ±1.64	87.36 ±1.44	84.91 ±1.97	90.34 ±1.26	94.04 ±1.14	**94.59** ±1.50
κ ×100	77.03 ±3.69	84.95 ±2.90	87.27 ±4.01	84.35 ±1.48	80.65 ±2.95	82.17 ±3.52	88.17 ±2.40	**89.32** ±2.52

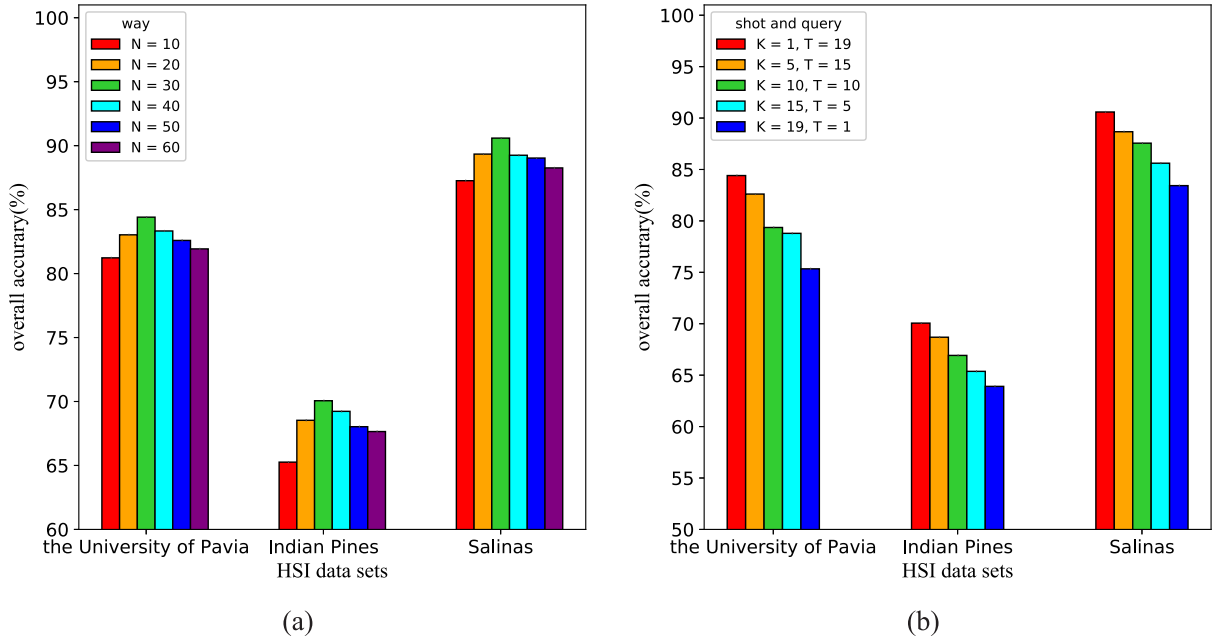


Fig. 7. OA under different episodic training settings. (a) OA corresponding to different N. (b) OA corresponding to different combinations of K and T.

and simultaneously extracts large-scale superpixel-level features and local pixel-level features of HSI. DCFSL adds a conditional adversarial domain adaptation strategy to FSL, which takes into account the domain shift between the source domains and the target domains; due to the additional strategies adopted by both CEGCN and DCFSL, satisfactory suboptimal classification performance can also be achieved on the three HSI datasets. However, classification of these two methods is still inferior to FSL-EGNN. In particular, the difference between

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TABLE X
EXECUTION TIME (S) FOR EACH DATASET WITH DIFFERENT METHODS, INCLUDING TRAINING TIME, FINE-TUNING TIME, AND TESTING TIME

Execution times		SVM-RBF	SSRN	CEGCN	DFSL+NN	DFSL+SVM	RN-FSC	DCFSL	FSL-EGNN
UP	Training time	0.17	1827.30	176.39	1672.92	1672.92	45350.94	3317.68	2281.16
	Fine-tuning time	N/A	N/A	N/A	N/A	N/A	151.88	N/A	18.75
	Testing time	0.34	8.20	5.30	1.27	0.56	27.12	1.09	47.64
IP	Training time	0.22	840.29	20.73	1672.92	1672.92	45350.94	3073.62	2281.16
	Fine-tuning time	N/A	N/A	N/A	N/A	N/A	50.77	N/A	30.82
	Testing time	0.19	3.86	0.68	0.50	0.97	10.46	0.51	14.02
SA	Training time	0.25	4274.20	88.80	1672.92	1672.92	45350.94	3159.49	2281.16
	Fine-tuning time	N/A	N/A	N/A	N/A	N/A	150.82	N/A	30.77
	Testing time	1.71	19.11	3.9	1.72	1.19	33.20	1.57	78.35

FSL-EGNN and RN-FSC is only that they use different networks to express the relationship between samples. However, FSL-EGNN can achieve better classification performance than RN-FSC on three HSI datasets, indicating that EGNN has more advantages in modeling the association between samples than the relational network.

2) As expected, when compared with other state-of-the-art methods, the proposed FSL-EGNN achieves a stable improvement in classification performance. Specifically, in terms of OA, the results of FSL-EGNN on the UP, IP, and Salinas datasets are 1.83%, 1.18%, and 1.05% higher than the results of the suboptimal method, respectively. In terms of AA, the results of FSL-EGNN on the UP, IP, and Salinas datasets are 2.18%, 2.98%, and 0.28% higher than the results of the suboptimal method, respectively. In terms of κ, the results of FSL-EGNN on the UP, IP, and Salinas datasets are 2.41%, 1.07%, and 1.15% higher than the results of the suboptimal method, respectively. Different from the general FSL method, FSL-EGNN models the relationship between pixels in HSIs through the edges in the graph structure and uses edge labels to explicitly express the intraclass similarity and interclass dissimilarity in HSIs in the process of FSL, which can effectively obtain the discriminative information and improve the classification performance with limited labeled samples.
3) FSL-EGNN can still obtain better classification performance for the misclassified classes of other state-of-the-art methods. For example, class 6 (Bare Soil) and class 8 (Self-Blocking Bricks) in the UP dataset, class 1 (Alfalfa), class 4 (Corn), class 7 (Grass-pasture-mowed), class 9 (Oats), and class 12 (Soybean-clean) in the IP dataset, and class 2 (Brocoli_green_weeds_2), class 14 (Lettuce_romaine_4wk), and class 15 (Vinyard_untrained) in the Salinas dataset. In particular, the classification results of class 8 (i.e., 87.32%) in the UP, class 1 (i.e., 98.26%) in the IP, and class 15 (i.e., 78.27%) in Salinas are 1.98%, 2.89%, and 2.89% higher than the suboptimal results (i.e., 85.34%, 95.37%, and 75.38%) of the same row, respectively. The above phenomenon indicates that FSL-EGNN can effectively overcome the obstacles of misclassification of classes, and the classification results are more stable and robust.

E. Qualitative Visual Comparisons

In addition to the quantitative comparisons in Tables VII–IX, the classification maps obtained by the eight different methods are used for qualitative visual comparisons. Figs. 8–10 reveal the classification maps of the UP, the IP, and the Salinas datasets under different methods, respectively. Moreover, the pseudocolor image and ground-truth map of each dataset are also provided; the pseudocolor image is obtained by selecting the 10th, 50th, and 100th bands of the corresponding HSI. When comparing the classification maps obtained by different methods in subpatterns (c)–(j) with the ground-truth map in subpattern (b), it can be found that the proposed FSL-EGNN can generate a reasonable classification map with less noise and more homogeneous regions, which can intuitively prove the superiority of FSL-EGNN.

F. Ablition Study

As mentioned in Section III, the proposed FSL-EGNN contains three critical parts to improve the classification performance, i.e., FEM, training phase on the source datasets, and fine-tuning phase on the target datasets. To reveal the contribution of the three components, ablation study was performed on three target datasets, including the UP, the IP, and the Salinas datasets. Specifically, we adopt "FSL-EGNN-V1," "FSL-EGNN-V2," and "FSL-EGNN-V3" to represent the reduced model by removing the FEM, the training phase on the source dataset, and the fine-tuning phase on the target dataset, respectively. It is worth noting that the three modules of EIM, EGNN, and LPM are connected end to end as a whole, which cannot be disassembled for the entire FSL-EGNN framework. Therefore, no ablation studies were conducted for EIM, EGNN, and LPM. In addition, to verify the superiority of 3-D convolution in FEM, we replace all the 3-D convolution layers of FEM in FSL-EGNN with 2-D convolution layers, and denote it as "FSL-EGNN-V4." The specific comparative results are exhibited in Fig. 11. Not surprisingly, comparing the classification performance of FSL-EGNN and FSL-EGNN-V1, FSL-EGNN-V2, and FSL-EGNN-V3, it can be seen that the lack of any components will inevitably hurt the classification results. Therefore, FEM, training phase on the source datasets, and fine-tuning phase on the target datasets work collaboratively to render satisfactory

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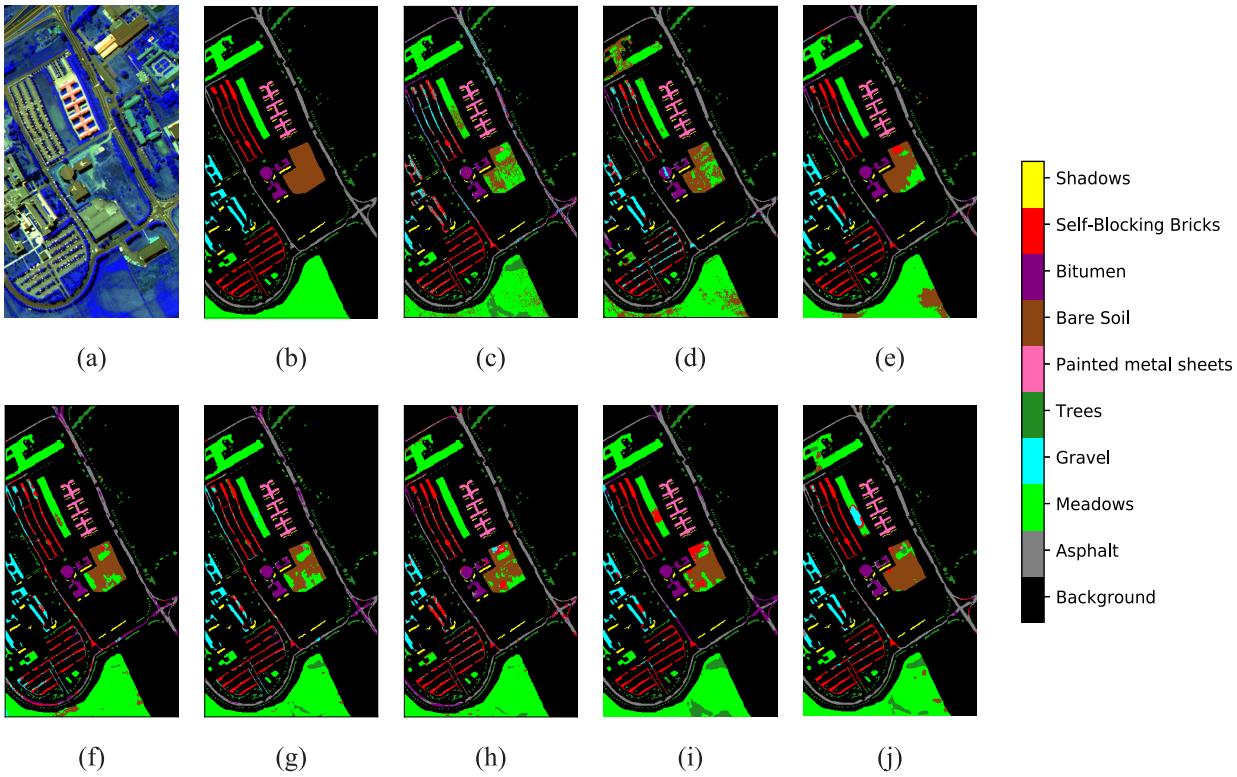


Fig. 8. Classification maps of different methods on the UP dataset. (a) Pseudocolor image (bands 10, 50, and 100). (b) Ground-truth map. (c) SVM-RBF (OA: 61.35%). (d) SSRN (OA: 76.26%). (e) CEGCN (OA: 80.22%). (f) DFSL + NN (OA: 74.23%). (g) DFSL + SVM (OA: 73.58%). (h) RN-FSC (OA: 77.58%). (i) DCFSL (OA: 83.65%). (j) FSL-EGNN (OA: 85.48%).

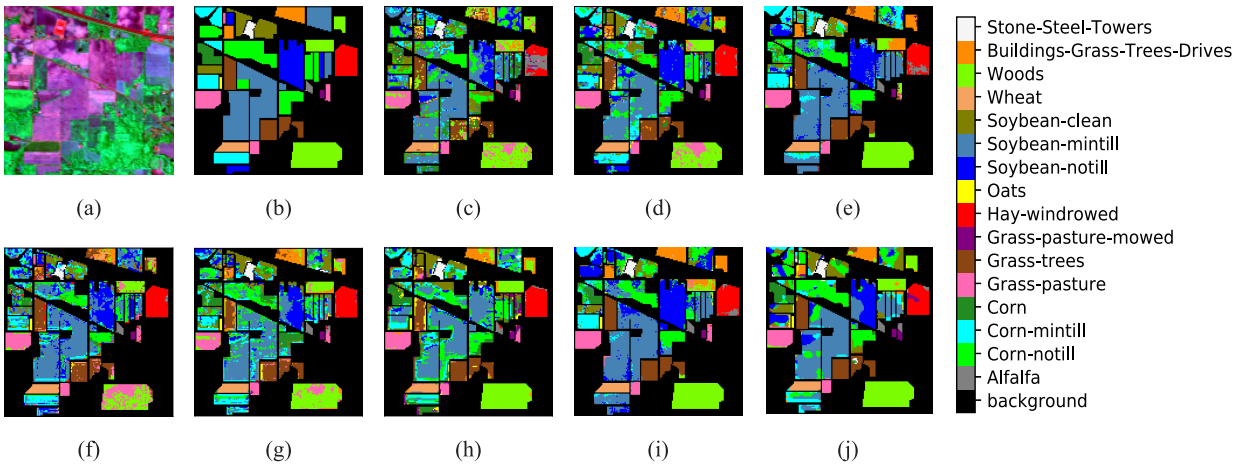


Fig. 9. Classification maps of different methods on the IP dataset. (a) Pseudocolor image (bands 10, 50, and 100). (b) Ground-truth map. (c) SVM-RBF (OA: 47.86%). (d) SSRN (OA: 61.36%). (e) CEGCN (OA: 65.95%). (f) DFSL + NN (OA: 60.04%). (g) DFSL + SVM (OA: 59.44%). (h) RN-FSC (OA: 52.49%). (i) DCFSL (OA: 66.81%). (j) FSL-EGNN (OA: 67.99%).

classification performance. Comparing FSL-EGNN and FSL-EGNN-V4, it can be seen that compared with 2-D convolution, 3-D convolution is more superior in FEM.

G. Impact of the Number of Labeled Samples

In this section, we investigate the classification performance of the proposed FSL-EGNN and other seven competitive methods with different training samples. To this end, we stipulate that the number of training samples obtained randomly from each class is 5, 10, 15, 20, 25, and 30, respectively, and report the average of the OA of ten trials on the UP, IP, and Salinas datasets, as shown in Fig. 12. From the results, it is not difficult to find that the classification performance of the proposed FSL-EGNN outperforms other competitors on the three datasets, which demonstrates that FSL-EGNN has better stability in the classification task with different numbers of training samples.

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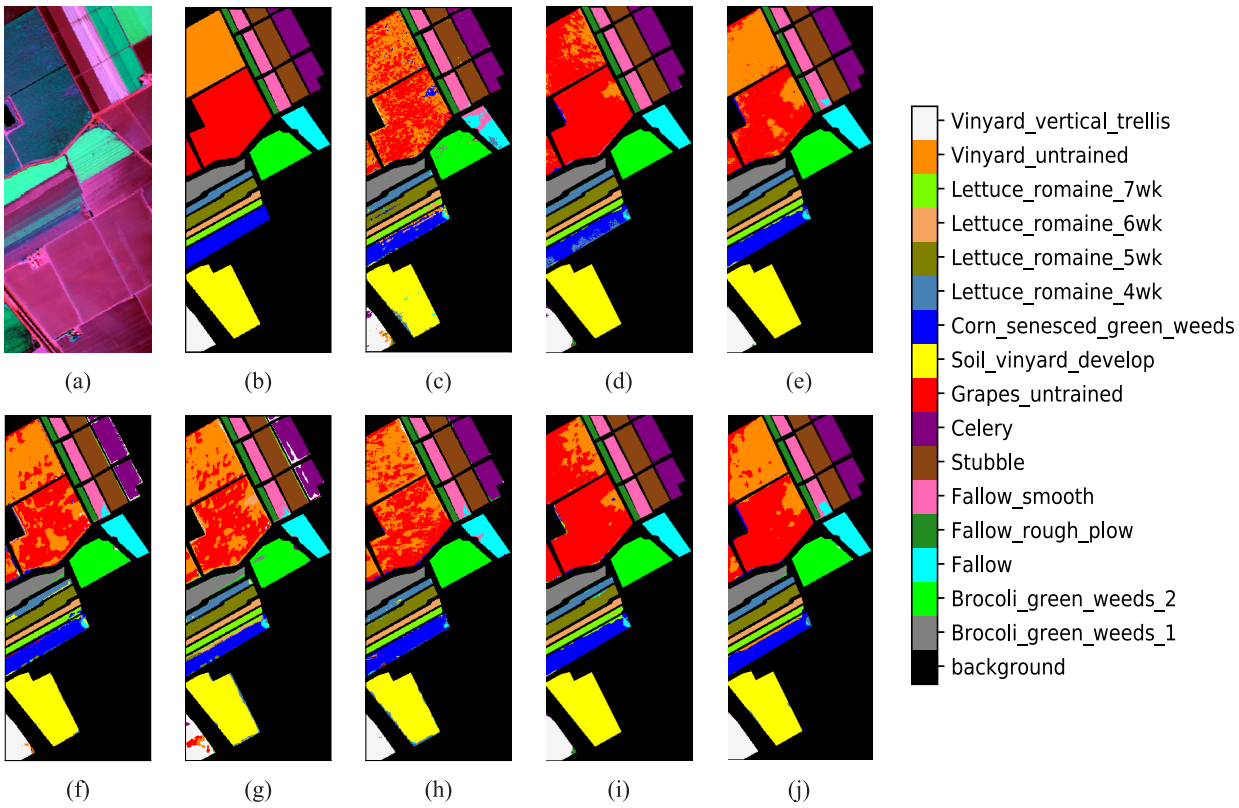


Fig. 10. Classification maps of different methods on the Salinas dataset. (a) Pseudocolor image (bands 10, 50, and 100). (b) Ground-truth map. (c) SVM-RBF (OA: 79.27%). (d) SSRN (OA: 86.39%). (e) CEGCN (OA: 88.55%). (f) DFSL + NN (OA: 85.85%). (g) DFSL + SVM (OA: 82.51%). (h) RN-FSC (OA: 83.89%). (i) DCFSL (OA: 89.34%). (j) FSL-EGNN (OA: 90.39%).

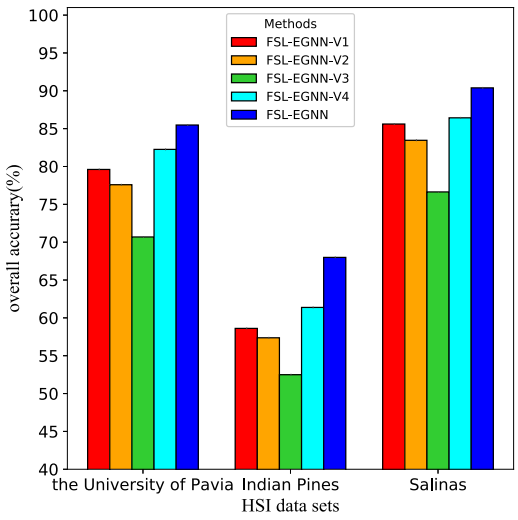


Fig. 11. Classification accuracy for the three HSI datasets with different methods.

H. Execution Time Analysis

As far as the procedure is concerned, the proposed FSL-EGNN includes three parts: the model training phase on the source dataset, the fine-tuning phase on the fine-tuning dataset, and the model testing phase on the testing dataset. Correspondingly, the execution time of FSL-EGNN consists of three parts: training time, fine-tuning time, and testing time. Table X shows the execution time of the aforementioned methods on the three target datasets (five training samples are randomly selected for each class). In detail, SVM-RBF, SSRN, CEGCN, DFSL + NN, DFSL + SVM, and DCFSL all have only model training phase and model testing phase, so the execution time of these methods does not include the fine-tuning time. Similar to FSL-EGNN, the execution time of RN-FSC consists of training time, fine-tuning time, and testing time. Compared with SVM-RBF, GCN, and SSRN, the FSL methods (i.e., DFSL + NN, DFSL + SVM, RN-FSC, FSL-EGNN) take longer training time, but SVM-RBF, GCN, and SSRN need to retrain the model for each classification task; on the contrary, the FSL methods only need to train the model once, and then can be extended to any datasets for classification, effectively improving the classification efficiency. Among the FSL methods, the training time of the proposed FSL-EGNN is between the DFSL methods and RN-FSC, the fine-tuning efficiency of FSL-EGNN is higher than that of RN-FSC, but FSL-EGNN takes more testing time, which is the disadvantage of FSL-EGNN. However, the proposed FSL-EGNN can better improve the classification performance of HSI in the case of small samples; the increase in testing time is acceptable.

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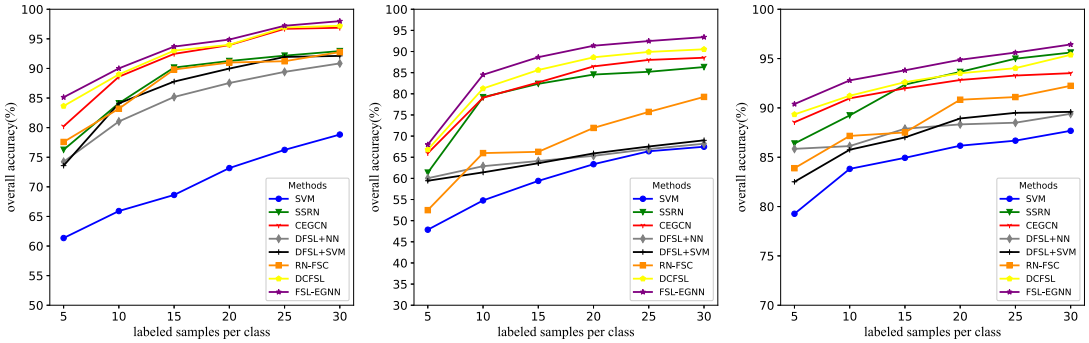


Fig. 12. Classification accuracy for the three HSI datasets with different sample numbers. (a) UP. (b) IP. (c) Salinas.

V. Conclusion

In this article, we have discovered FSL-EGNN, which is the first attempt to explicitly quantify the associations between pixels by exploiting EGNN in HSI FSC. Different from the general FSL methods, FSL-EGNN adopts graphs to characterize the association between HSI pixels and iteratively updates the edge labels in the graph to infer the association between query nodes and support nodes in FSL. Specifically, in the process of EGNN, the edge features and node features are alternately updated, and the obtained edge labels can explicitly characterize the intracluster similarity and intercluster dissimilarity between nodes. Finally, the existing labels of support nodes and edge labels are used together to obtain the predicted labels of the query nodes. The edge loss is used to update the parameters of EGNN with episodic training. The performance of FSL-EGNN has been verified on three benchmark HSIs compared with the state-of-the-art methods, and the experimental results have confirmed the superiority of the proposed FSL-EGNN.

FSL-EGNN exhibits a superior classification performance to other methods under the condition of small samples. However, the execution efficiency and classification accuracy of FSL-EGNN still need to be improved. Further work will focus on collecting more representative source datasets and embedding lighter weight GNNs in FSL.

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References

[1] B. Liu, X. Yu, P. Zhang, A. Yu, Q. Fu, and X. Wei, "Supervised deep feature extraction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1909–1921, Apr. 2018.

[2] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral–spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2017.

[3] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Apr. 2019.

[4] C. Niu, K. Tan, X. Jia, and X. Wang, "Deep learning based regression for optically inactive inland water quality parameter estimation using airborne hyperspectral imagery," *Environ. Pollut.*, vol. 286, Oct. 2021, Art. no. 117534.

[5] Y.-Q. Wan, Y.-H. Fan, and M.-S. Jin, "Application of hyperspectral remote sensing for supplementary investigation of polymetallic deposits in huaniushan ore region, northwestern China," *Sci. Rep.*, vol. 11, no. 1, p. 440, Dec. 2021.

[6] F. Zhang *et al.*, "Hyperspectral imaging combined with convolutional neural network for outdoor detection of potato diseases," in *Proc. 6th Int. Symp. Comput. Inf. Process. Technol. (ISCIPT)*, Jun. 2021, pp. 846–850.

[7] S. Meng, X. Wang, X. Hu, C. Luo, and Y. Zhong, "Deep learning-based crop mapping in the cloudy season using one-shot hyperspectral satellite imagery," *Comput. Electron. Agricult.*, vol. 186, Jul. 2021, Art. no. 106188.

[8] P. Ghamisi *et al.*, "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[9] F. Xue, F. Tan, Z. Ye, J. Chen, and Y. Wei, "Spectral–spatial classification of hyperspectral image using improved functional principal component analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[10] Q. Hou, Y. Wang, L. Jing, and H. Chen, "Linear discriminant analysis based on kernel-based possibilistic C-means for hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 8, pp. 1259–1263, Aug. 2019.

[11] J. Huang, K. Liu, M. Xu, M. Perc, and X. Li, "Background purification framework with extended morphological attribute profile for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8113–8124, 2021.

[12] W. Li, C. Chen, H. Su, and Q. Du, "Local binary patterns and extreme learning machine for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3681–3693, Jul. 2015.

[13] Y. Zhang, K. Liu, Y. Dong, K. Wu, and X. Hu, "Semisupervised classification based on SLIC segmentation for hyperspectral image," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 8, pp. 1440–1444, Aug. 2020.

[14] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[15] L. Samaniego, A. Bardossy, and K. Schulz, "Supervised classification of remotely sensed imagery using a modified *k*-NN technique," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2112–2125, Jun. 2008.

[16] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[17] Q. Liu, L. Xiao, J. Yang, and J. C.-W. Chan, "Content-guided convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6124–6137, Sep. 2020.

[18] Y. Chen, Y. Wang, Y. Gu, X. He, P. Ghamisi, and X. Jia, "Deep learning ensemble for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1882–1897, Jun. 2019.

[19] B. Liu, A. Yu, X. Yu, R. Wang, K. Gao, and W. Guo, "Deep multiview learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7758–7772, Sep. 2021.

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[20] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[21] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[22] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Jul. 2016.

[23] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[24] S. Hao, W. Wang, and M. Salzmann, "Geometry-aware deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2448–2460, Mar. 2021.

[25] J. Yue, W. Zhao, S. Mao, and H. Liu, "Spectral–spatial classification of hyperspectral images using deep convolutional neural networks," *Remote Sens. Lett.*, vol. 6, no. 6, pp. 468–477, May 2015.

[26] J. Yue, S. Mao, and M. Li, "A deep learning framework for hyperspectral image classification using spatial pyramid pooling," *Remote Sens. Lett.*, vol. 7, no. 9, pp. 875–884, 2016.

[27] Y. Chen, L. Zhu, P. Ghamisi, X. Jia, G. Li, and L. Tang, "Hyperspectral images classification with Gabor filtering and convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2355–2359, Dec. 2017.

[28] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[29] M. E. Paoletti, J. M. Haut, R. Fernandez-Beltran, J. Plaza, A. J. Plaza, and F. Pla, "Deep pyramidal residual networks for spectral–spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 740–754, Feb. 2019.

[30] C. Zhang, G. Li, and S. Du, "Multi-scale dense networks for hyperspectral remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9201–9222, Nov. 2019.

[31] B. Liu, X. Yu, P. Zhang, X. Tan, R. Wang, and L. Zhi, "Spectral–spatial classification of hyperspectral image using three-dimensional convolution network," *J. Appl. Remote Sens.*, vol. 12, no. 1, pp. 1–18, 2018.

[32] S. Ghaderizadeh, D. Abbasi-Moghadam, A. Sharifi, N. Zhao, and A. Tariq, "Hyperspectral image classification using a hybrid 3D-2D convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7570–7588, 2021.

[33] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[34] Z. Zhong, J. Li, D. A. Clausi, and A. Wong, "Generative adversarial networks and conditional random fields for hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 50, no. 7, pp. 3318–3329, Jul. 2020.

[35] R. Hang, F. Zhou, Q. Liu, and P. Ghamisi, "Classification of hyperspectral images via multitask generative adversarial networks," *IEEE Trans. Geoence Remote Sens.*, vol. 59, no. 2, pp. 1424–1436, Feb. 2020.

[36] J. Wang, F. Gao, J. Dong, and Q. Du, "Adaptive DropBlock-enhanced generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5040–5053, Jun. 2021.

[37] M. E. Paoletti *et al.*, "Capsule networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2145–2160, Apr. 2019.

[38] J. Wang, S. Guo, R. Huang, L. Li, X. Zhang, and L. Jiao, "Dual-channel capsule generation adversarial network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[39] R. Lei *et al.*, "Hyperspectral remote sensing image classification using deep convolutional capsule network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8297–8315, 2021.

[40] X. He and Y. Chen, "Optimized input for CNN-based hyperspectral image classification using spatial transformer network," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 12, pp. 1884–1888, Dec. 2019.

[41] Z. Zhong, Y. Li, L. Ma, J. Li, and W.-S. Zheng, "Spectral–spatial transformer network for hyperspectral image classification: A factorized architecture search framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[42] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[43] L. Mou, P. Ghamisi, and X. X. Zhu, "Unsupervised spectral–spatial feature learning via deep residual Conv–Deconv network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 391–406, Jan. 2018.

[44] W. Wang, D. Shuguang, J. Zhongmin, and S. Liujie, "A fast dense spectral–spatial convolution network framework for hyperspectral images classification," *Remote Sens.*, vol. 10, no. 7, p. 1068, Jul. 2018.

[45] Z. Li *et al.*, "Deep multilayer fusion dense network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1258–1270, 2020.

[46] Y. Tong *et al.*, "Multi-scale feature extraction and total variation based fusion method for HSI and lidar data classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 5433–5436.

[47] W. Ma, Q. Yang, Y. Wu, W. Zhao, and X. Zhang, "Double-branch multi-attention mechanism network for hyperspectral image classification," *Remote Sens.*, vol. 11, no. 11, p. 1307, 2019.

[48] R. LI, S. Zheng, C. Duan, Y. Yang, and X. Wang, "Classification of hyperspectral image based on double-branch dual-attention mechanism network," 02 2020.

[49] Z. Xue, X. Yu, B. Liu, X. Tan, and X. Wei, "HResNetAM: Hierarchical residual network with attention mechanism for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3566–3580, 2021.

[50] D. Hong, J. Chanussot, N. Yokoya, U. Heiden, W. Heldens, and X. X. Zhu, "WU-Net: A weakly-supervised unmixing network for remotely sensed hyperspectral imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 373–376.

[51] E. Moliner, L. S. Romero, and V. Vilaplana, "Weakly supervised semantic segmentation for remote sensing hyperspectral imaging," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 2273–2277.

[52] S. Jia, B. Zhao, L. Tang, F. Feng, and W. Wang, "Spectral–spatial classification of hyperspectral remote sensing image based on capsule network," *J. Eng.*, vol. 2019, no. 21, pp. 7352–7355, Nov. 2019.

[53] X. Jiang, W. Liu, Y. Zhang, J. Liu, S. Li, and J. Lin, "Spectral–spatial hyperspectral image classification using dual-channel capsule networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 6, pp. 1094–1098, Jun. 2021.

[54] B. Liu, X. Yu, P. Zhang, X. Tan, A. Yu, and Z. Xue, "A semi-supervised convolutional neural network for hyperspectral image classification," *Remote Sens. Lett.*, vol. 8, no. 9, pp. 839–848, Sep. 2017.

[55] S. Mei, J. Ji, Y. Geng, Z. Zhang, X. Li, and Q. Du, "Unsupervised spatial–spectral feature learning by 3D convolutional autoencoder for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6808–6820, Apr. 2019.

[56] S. Jia *et al.*, "A semisupervised Siamese network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[57] A. Qin, Z. Shang, J. Tian, Y. Wang, T. Zhang, and Y. Tang, "Spectral–spatial graph convolutional networks for semisupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 241–245, Feb. 2019.

[58] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2020.

[59] T. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. 5th Int. Conf. Learn. Represent. (ICLR)*. Toulon, France, Apr. 2017, pp. 24–26.

[60] Z. U. O. Xibing *et al.*, "Graph convolutional network method for small sample classification of hyperspectral images," *Acta Geodaetica et Cartographica Sinica*, vol. 50, no. 10, p. 1358, 2021.

[61] K. Xu, Y. Zhao, L. Zhang, C. Gao, and H. Huang, "Spectral–spatial residual graph attention network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[62] Y. Ding, X. Zhao, Z. Zhang, W. Cai, and N. Yang, "Multiscale graph sample and aggregate network with context-aware learning for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4561–4572, 2021.

[63] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

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[64] K. Gao, B. Liu, X. Yu, J. Qin, P. Zhang, and X. Tan, "Deep relation network for hyperspectral image few-shot classification," *Remote Sens.*, vol. 12, no. 6, p. 923, Mar. 2020.
[65] X. Tong, J. Yin, B. Han, and H. Qv, "Few-shot learning with attention-weighted graph convolutional networks for hyperspectral image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 1686–1690.
[66] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.
[67] K. Sun *et al.*, "A robust and efficient band selection method using graph representation for hyperspectral imagery," *Int. J. Remote Sens.*, vol. 37, no. 20, pp. 4874–4889, 2016.
[68] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel- and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.



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